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Roll No

CE-5005(2)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Construction Planning and Management

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What are the main advantages of Activity on the Node Over Activity on the Arrow networks? 7
- b) Draw a PERT network for the following project: 7
 - i) 'A' is the first or start event and 'K' is the end event
 - ii) 'J' is successor event of 'F'
 - iii) 'C' and 'D' are successor event of B
 - iv) 'D' is a preceding event of 'G'
 - v) 'E' and 'F' occur after event 'C'
 - vi) 'E' precedes 'F'
 - vii) 'C' restrains the occurrence of 'G' and 'G' precedes 'H'
 - viii) 'H' precedes 'J'
 - ix) 'F' restrains the occurrence of 'H'
 - x) 'K' succeeds event 'J'

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PTO

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2. Information of the activities required for a medium-size Civil Engineering project is as follows: 14

Activity	A	B	C	D	E	F	G	H	I	J	K
Node No.	1-2	1-3	1-4	2-5	3-5	3-6	3-7	4-6	5-7	6-8	7-8
Duration (in months)	2	7	8	3	6	10	4	6	2	5	6

Draw the network and calculate the following:

- i) Earliest and latest event time
 - ii) Earliest and latest start time
 - iii) Earliest and latest finish time
 - iv) Float
 - v) Critical path
3. Define the following: 14
- a) Depreciation cost and Investment cost
 - b) Dozers
 - c) Drag line
 - d) Scraper
 - e) Sheepsfoot Roller
 - f) Batch Mixer
 - g) Chain Hoists
4. a) What are the different types of tender? Explain notice inviting tender. 7
- b) What do you mean by: 7
- i) Compensation for delay in completion
 - ii) Liquidated Damages and Unliquidated Damages
 - iii) Extension of Time
 - iv) Termination of contract

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5. a) What is Measurement Book (M.B.)? What are the common irregularities found in Measurement Books? 7
b) Write details specification of 1st class brickwork. 7
6. a) Explain briefly the types of organisation. 7
b) What is Trade Union? Discuss 'Trade Union Act'. 7
7. a) Describe briefly the major ways in which a CPM network differs from a PERT network. In what circumstances these network are used? 10
b) Define 'Specifications'. What is the necessity of drawing up building specifications? 4
8. Write short note on following: 14
 - i) Job layout
 - ii) Schedule of rates
 - iii) Lumpsum contract
 - iv) Earnest Money and Security Money
